



Answer the following questions:

Question – 1:

Put True (✓) or False (×) signs for the following expressions (with Correction if found): [10 Marks]

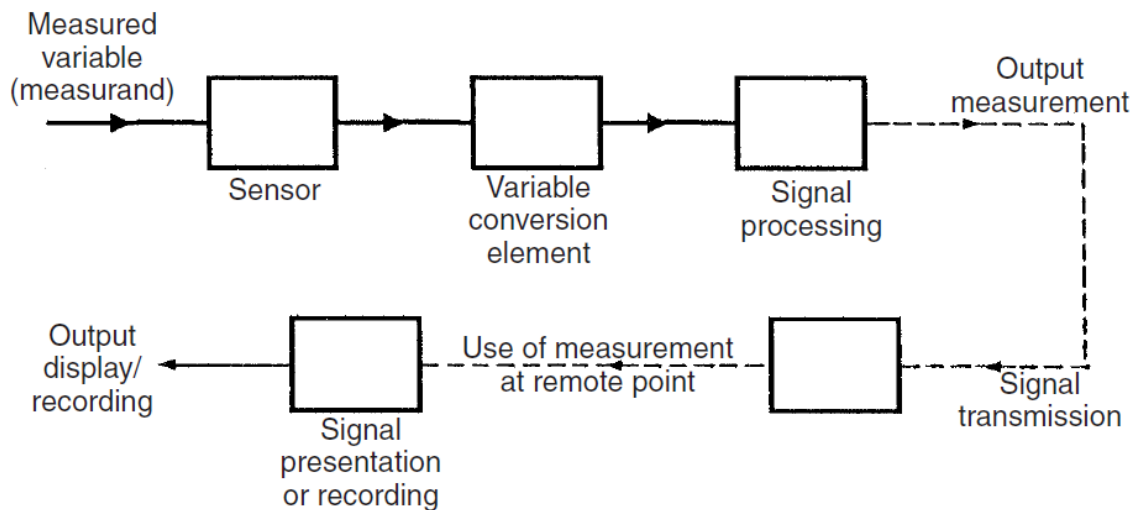
1. Threshold (**resolution**) is a lower limit on the magnitude of the change in the input measured quantity that produces an observable change in the instrument output. ×
2. A car speedometer typically has a resolution (**Threshold**) of about 15 km/h. This means that, if the vehicle starts from rest and accelerates, no output reading is observed on the speedometer until the speed reaches 15 km/h. ×
3. In the active (**Passive**) instrument the external power supply is not need. ×
4. Measurement system is useful in feedback control systems only. ×
5. An analog instrument gives an output that varies continuously as quantity being measured. ✓
6. The threshold (**range or span**) of an instrument defines the minimum and maximum values of a quantity that the instrument is designed to measure. ×
7. Tolerance (**Precision**) is a term that describes an instrument's degree of freedom from random errors. ×
8. The precision (**accuracy**) of an instrument is a measure of how close the output reading of the instrument is to the correct value. ×
9. Reproducibility (**Repeatability**) describes the closeness of output readings when the same input is applied repetitively over a short period of time, with the same measurement conditions. ×
10. If a large number of readings are taken of the same quantity by a high precision instrument, then the spread of readings will be very small. ✓

Question – 2:

[10 Marks]

1. Draw the block diagram of measurement system.

[2.5 marks]



2. State the disadvantages of the conversion from A/D.

[2.5 marks]

1. Firstly, the A/D converter adds a significant cost to the system.
2. Secondly, a finite time is involved in the process of converting an analogue signal to a digital quantity, and this time can be critical in the control of fast processes where the accuracy of control depends on the speed of the controlling computer.

3. A voltmeter, having a sensitivity of $1,000\Omega /V$, reads 100 V on its 150-V scale when connected across an unknown resistor in series with a millimeter. When the millimeter reads 5 mA, Calculate:

1. Apparent resistance of the unknown resistor,
2. Actual resistance of the unknown resistor,
3. Error due to the loading effect of the voltmeter.

[5 marks]

Solution

1. The total circuit resistance equals

$$R_T = \frac{V_T}{I_T} = \frac{100 \text{ V}}{5 \text{ mA}} = 20 \text{ k}\Omega$$

Neglecting the resistance of the milliammeter, the value of the unknown resistor is $R_X = 20 \text{ k}$.

2. The voltmeter resistance equals

$$R_V = 1000 \frac{\Omega}{V} \times 150 \text{ V} = 150 \text{ k}\Omega$$

Since the voltmeter is in parallel with the unknown resistance, we can write

$$R_X = \frac{R_T R_V}{R_V - R_T} = \frac{20 \times 150}{130} = 23.05 \text{ k}\Omega$$

3. Error

$$\% \text{ Error} = \frac{\text{actual} - \text{apparent}}{\text{actual}} \times 100\% = \frac{23.05 - 20}{23.05} \times 100\% = 13.23\%$$

With best wishes

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